This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

JustRAIGS: Justified Referral in AI Glaucoma Screening Challenge

Yeganeh Madadi, *Member, IEEE*, Hina Raja, Koenraad A. Vermeer, Hans G. Lemij, Xiaoqin Huang, Eunjin Kim, Seunghoon Lee, Gitaek Kwon, Hyunwoo Kim, Jaeyoung Kim, Adrian Galdran, Miguel A. Gonza´lez Ballester, Dan Presil, Kristhian Aguilar, Victor Cavalcante, Celso Carvalho, Waldir Sabino, Mateus Oliveira, Hui Lin, Charilaos Apostolidis, Aggelos K. Katsaggelos, Tomasz Kubrak, A´. Casado-Garc´ıa, J. Heras, M. Ortega, L. Ramos, Philippe Zhang, Yihao Li, Jing Zhang, Weili Jiang, Pierre-Henri Conze, Mathieu Lamard, Gwenole´ Quellec, Mostafa El Habib Daho, Madukuri Shaurya, Anumeha Varma, Monika Agrawal, and Siamak Yousefi, *Senior Member, IEEE*

***Abstract*— A major contributor to permanent vision loss is glaucoma. Early diagnosis is crucial for preventing vision loss due to glaucoma, making glaucoma screening essential. A more affordable method of glaucoma screening can be achieved by applying artificial intelligence to evaluate color fundus photographs (CFPs). We present the Justified Referral in AI Glaucoma Screening (JustRAIGS) challenge to further develop these AI algorithms for glaucoma screening and to assess their efficacy. To support this challenge, we have generated a distinctive big dataset containing more than 110,000 meticulously labeled CFPs obtained from approximately 60,000 patients and 500 distinct screening centers in the USA. Our objective is to assess the practicality of creating advanced and dependable AI systems that can take a CFP as input and produce the probability of referable glaucoma, as well as outputs for glaucoma justification by integrating both binary and multi-label classification tasks. This paper presents the evaluation of solutions provided by nine teams, recognizing the team with the highest level of performance. The highest achieved score of sensitivity at a specificity level of 95% was 85%, and the highest achieved score of Hamming losses average was 0.13. Additionally, we test the top three participants' algorithms on an external dataset to validate the performance and generalization of these models. The outcomes of this research can offer valuable insights into the development of intelligent systems for detecting glaucoma. Ultimately, findings can aid in the early detection and treatment of glaucoma patients, hence decreasing preventable vision impairment and blindness caused by glaucoma.**

***Index Terms*—Artificial intelligence, justified referral glaucoma screening, classification task.**

I. Introduction

WORLDWIDE, glaucoma is a leading cause of permanent visual loss and blindness. It gradually

This work was supported in part by the National Institutes of Health Grant EY033005.

Corresponding authors: Yeganeh Madadi (e-mail: madadiy@appstate.edu), Siamak Yousefi (e-mail: siamak.yousefi@uthsc.edu).
Yeganeh Madadi is with the Department of Computer Science, Appalachian State University, Boone, NC, USA.
Hina Raja is with the Department of Mathematics and Computer Science, Fisk University, Nashville, TN, USA.
Koenraad A. Vermeer and Hans G. Lemij are with the Rotterdam Ophthalmic Institute, Rotterdam Eye Hospital, Rotterdam, Netherlands.
Xiaoqin Huang is with the Division of Intramural Research, National Library of Medicine, National Institutes of Health, MD, USA.
Eunjin Kim, Seunghoon Lee, Gitaek Kwon, and Hyunwoo Kim are with the VUNO Inc., Seoul, South Korea.
Jaeyoung Kim is with Teamreboott Inc., Busan, South Korea.
Adrian Galdran is with Tecnalia Research and Innovation, Spain.
Miguel A. Gonza´lez Ballester is with BCN Medtech, Department of Information and Communication Technologies, Universitat Pompeu Fabra, Barcelona, Spain. ICREA, Barcelona, Spain.
Dan Presil is with Ben-Gurion University of the Negev, Israel.
Kristhian Aguilar, Victor Cavalcante, Celso Carvalho, Waldir Sabino, and Mateus Oliveira are with the Faculty of Technology, Federal University of Amazonas, Amazonas, Brazil.
Hui Lin, Charilaos Apostolidis, and Aggelos K. Katsaggelos are with the Department of Electrical and Computer Engineering, Northwestern University, Evanston, IL, USA.
Tomasz Kubrak is with Vrije University, Netherlands.
A´. Casado-Garc´ıa, J. Heras is with Grupo de Informa´tica Departamento de Matema´ticas y Computacio´n Universidad de La Rioja, Spain.
M. Ortega, L. Ramos is with Grupo VARPA Instituto de Investigacio´n Biome´dica Universidade da Corun˜a, Spain.
Philippe Zhang is with LaTIM UMR 1101, Inserm, Brest, France. University of Western Brittany, Brest, France. Evolucare Technologies, Villers-Bretonneux, France.
Yihao Li is with United Imaging Healthcare, Shanghai, China.
Jing Zhang, Mathieu Lamard, and Mostafa El Habib Daho are with LaTIM UMR 1101, Inserm, Brest, France. University of Western Brittany, Brest, France.
Weili Jiang is with the College of Computer Science, Sichuan University, Chengdu, China.
Pierre-Henri Conze is with LaTIM UMR 1101, Inserm, Brest, France. IMT Atlantique, Brest, France.
Gwenole´ Quellec is with LaTIM UMR 1101, Inserm, Brest, France.
Madukuri Shaurya, Anumeha Varma, and Monika Agrawal are with Indian Institute of Technology Delhi, Hauz Khas, New Delhi, Delhi, India.
Siamak Yousefi is with the Department of Ophthalmology, University of Tennessee Health Science Center, Memphis, TN, USA.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

damages the visual field by affecting the optic nerve, which links the eye to the brain. Glaucoma is expected to cause varied degrees of visual impairment in about 110 million individuals by 2040 [1]. Ten percent of those people will become blind in both eyes, and twenty-four percent in one eye [2]. Glaucoma is a common cause of blindness, and it is typically not diagnosed until the disease has progressed significantly [3]. The damage cannot be repaired by current glaucoma treatments; at most, they can only reduce or stop the condition's progression [4]. To lessen the impact of the condition, screening systems should be put in place to detect individuals at an early stage so that they can receive treatment. Automated detection of glaucoma in color fundus photographs (CFPs)[1] [5-9] suggests that artificial intelligence (AI) might pave the way for these systems' cost-effective adoption [10].

Justified referral in glaucoma screening refers to the appropriate recommendation for patients to undergo further specialist evaluation based on clear clinical indicators of potential glaucoma, and it is crucial for ensuring timely treatment while avoiding unnecessary strain on healthcare resources. However, achieving justified referrals is fraught with several issues that complicate effective screening and treatment of the disease. An important problem is the elevated occurrence of false-positive and false-negative results, which can arise due to inconsistencies in diagnostic criteria and the subjective analysis of test results, such as assessments of visual fields and evaluations of the optic nerve [11]. This variability often leads to inconsistent referral decisions, with some patients being referred unnecessarily, thus increasing the burden on specialized clinics, while others who require timely care may be overlooked [12]. While one might argue that using black-box models in screening is acceptable since a medical doctor will ultimately diagnose the patient, relying on opaque systems can hinder clinicians' trust and limit the interpretability necessary for effective decision-making. Inconsistent referral procedures can be further exacerbated by variations in the quality and availability of diagnostic tools, as well as differing levels of experience among practitioners. These issues emphasize the necessity for more robust, unbiased, interpretable, and standardized diagnostic techniques to guarantee prompt and precise referrals in glaucoma screening [13].

To address the problems associated with justified referral in glaucoma screening, we initiated the Justified Referral in AI Glaucoma Screening (JustRAIGS) challenge. The challenge required participants to analyze CFPs and categorize each image into one of two classes: "referable glaucoma" (RG)[2] or "no referable glaucoma" (NRG)[3]. The "referable glaucoma" is a term used to describe eyes that show signs or features in the fundus images that indicate the presence of visual field loss due to glaucoma. These signs or features are significant enough to warrant additional examination or referral to a specialist. Conversely, "no referable glaucoma" pertains to instances where the fundus imaging does not exhibit substantial signs of glaucoma and therefore does not necessitate prompt referral. Glaucoma in an early stage that does not yet cause visual field damage would likewise be categorized as "no referable glaucoma". If the image is classified as “referable glaucoma”, participants will be required to do multi-label classification for ten additional features associated with glaucoma, in addition to the referable glaucoma classification. These features refer to distinct traits or irregularities that can be observed in the fundus images of individuals with glaucoma. The multi-label classification challenge entails giving appropriate labels to each fundus image by considering the presence or absence of specified features. These additional features offer more comprehensive information regarding the specific attributes detected in these fundus images. Our objective is to assess the practicality of creating advanced and dependable AI systems that can take a CFP as input and produce the probability of referable glaucoma, as well as outputs for glaucoma justification. This will be achieved by combining binary classification (referable vs. no referable glaucoma) and multi-label classification (for ten additional features). The outcomes of these classification tasks can offer valuable understanding into the creation of automated systems or algorithms for the detection of glaucoma, ultimately aiding in the early recognition and treatment of glaucoma patients, thus decreasing preventable visual impairment and blindness caused by glaucoma.

The JustRAIGS engaged in the challenge program of the International Symposium on Biomedical Imaging (ISBI) 2024. The JustRAIGS challenge website was developed on the Grand Challenge platform[4]. It is the continuation of our

[1] Color Fundus Photographs (CFPs) are high-resolution images of the interior surface of the eye, including the retina, optic disc, macula, and posterior pole.

[2] Referable Glaucoma (RG) refers to cases of glaucoma or suspected glaucoma where the signs observed in an eye examination or imaging study are significant enough to warrant referral to a specialist for further evaluation and potential treatment.

[3] No Referable Glaucoma (NRG) means that the optic nerve and surrounding retinal structures appear normal or do not show sufficient signs of glaucomatous damage to justify further evaluation. This may include very early cases or healthy eyes.

[4] https://justraigs.grand-challenge.org

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

TABLE I

STATISTICS OF THE JUSTRAIGS DATASET [14]. # = NUMBER OF. CFPS = COLOR FUNDUS PHOTOGRAPHS. RG = REFERABLE GLAUCOMA. NRG = NO REFERABLE GLAUCOMA.

			Full dataset	Training dataset	Test dataset
# CFPs / # patients			111,178 / 61,919	101,442 / 55,736	9,736 / 6,183
Prevalence (# CFPs / # patients)		RG	4,872 / 3,531	3,270 / 2,336	1,602 / 1,195
		NRG	106,306 / 58,388	98,172 / 53,400	8,134 / 4,988
Age at encounter (years; mean ± std. dev.)		RG	63.3 ± 10.5	63.0 ± 10.5	63.9 ± 10.7
		NRG	56.5 ± 10.2	56.5 ± 10.1	57.5 ± 10.4
Cameras (# CFPs (% within dataset) / # patients)	Canon	CR1	11,462 (10.2%) / 6,013	10,274 (10.1%) / 5,422	1,188 (10.5%) / 591
	Canon	CR2	10,523 (9.3%) / 5,538	9,179 (9.0%) / 4,866	1,344 (11.9%) / 672
	Canon	DGI	10,644 (9.4%) / 5,690	9,581 (9.4%) / 5,145	1,063 (9.4%) / 545
	Optovue	iCam 100	29,108 (25.8%) / 16,166	26,480 (26.1%) / 14,742	2,628 (23.3%) / 1,424
	Topcon	NW200	3,109 (2.8%) / 1,588	2,888 (2.8%) / 1,478	221 (2.0%) / 110
	Topcon	NW400	22,519 (20.0%) / 11,736	20,557 (20.3%) / 10,737	1,962 (17.4%) / 999
	Centervue	DRS	1,805 (1.6%) / 988	1,598 (1.6%) / 879	207 (1.8%) / 109
	Nidek	AFC300	61 (0.1%) / 31	53 (0.1%) / 27	8 (0.1%) / 4
	Crystalvue	NFC-700	8 (0.0%) / 4	6 (0.0%) / 3	2 (0.0%) / 1
	Unknown	N/A	23,493 (20.8%) / 12,323	20,826 (20.5%) / 10,981	2,667 (23.6%) / 1,342

previous challenge (AIROGS) [14] with new goals to expand the clinical scope through a dual-task design. While AIROGS focused solely on binary glaucoma classification (referable vs. no referable glaucoma), JustRAIGS introduces an additional dimension: multi-label justification. In JustRAIGS, participants are required to design algorithms that not only perform binary classification but also identify ten clinically relevant glaucomatous features for each referred case. This new justification task enhances the interpretability of algorithmic decisions and better reflects real-world clinical workflows, where referrals are expected to be supported by specific diagnostic findings. Our dataset is much larger than any other publicly accessible CFP dataset with justified glaucomatous labels. It consists of 111,178 CFPs, which is larger than other publicly available datasets that comprise CFPs with glaucoma.

The sizes of these other datasets range from 22 to 2,000 CFPs [15-22]. The JustRAIGS dataset is particularly diverse, originating from 500 screening centers across the USA and collected using a variety of cameras. In addition, our dataset is the first publicly available CFP dataset to include annotations for ten glaucomatous features, making it a valuable resource for glaucoma research [23]. Furthermore, the JustRAIGS challenge stands out as the inaugural competition that places significant emphasis on justified referral glaucoma screening and eliminates the potential for human interference in the production of test set outcomes, hence minimizing the risk of dishonest behavior. Finally, it significantly enhances the ability to replicate results, enabling anyone to utilize the trained algorithms that were presented and implement them on fresh data inside a cloud-based setting.

II. DATASETS

A. The JustRAIGS Dataset

The JustRAIGS dataset includes 111,178 CFPs from 61,919 people across 500 sites, with diverse ethnicities. EyePACS LLC of Santa Cruz, California, USA, provided the imaging data. The labels were provided by the Rotterdam Ophthalmic Institute at Rotterdam Eye Hospital in Rotterdam, Netherlands. All graders were taught and then selected for the CFP grading task through the European Optic Disc Assessment Trial (EODAT) [24], which included 110 stereoscopic optic nerve images in which all glaucomatous eyes demonstrated consistent visual field abnormalities on standard automated perimetry. 90 experienced ophthalmologists and optometrists were evaluated, and only those with at least 85% total accuracy and 92% specificity were chosen to label images for the current study. Eventually, 30 of 90 candidates passed. The graders underwent both initial evaluation and periodic monitoring throughout the grading process. For more information about this dataset, please refer to [14, 23].

In the annotation process, each eye was graded by two distinct graders, neither of whom knew the other's identity. If two graders agreed on a CFP label, it became the final label. In cases where disagreement occurred, the image was adjudicated by one of two senior glaucoma specialists who had previously achieved ≥95% accuracy on the EODAT. The final label was thereafter determined by his/her discernment. To monitor grading quality, periodic inter-rater agreement was checked, and reassessment procedures were performed. This multi-step grading protocol ensured labeling consistency and minimized inter-observer variability. If graders observed glaucomatous symptoms that they

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

suspected had been associated with visual field deficiencies on standard automated perimetry, they were to mark the patient as RG. There were the following signs available for selection: "appearance neuroretinal rim inferiorly", "appearance neuroretinal rim superiorly", "baring of the circumlinear vessel inferiorly", "baring of the circumlinear vessel superiorly", "disc hemorrhage(s)", "retinal nerve fiber layer defect inferiorly, "retinal nerve fiber layer defect superiorly", "laminar dots" and "large cup", "nasalization (nasal displacement) of the vessel trunk". These ten glaucoma-related features were selected through a collaborative process involving experienced glaucoma specialists and other expert annotators at the Rotterdam Ophthalmic Institute, based on clinical consensus and established guidelines from prior diagnostic studies. The features were chosen for their known relevance in evaluating glaucomatous optic neuropathy from color fundus photography and their frequent use in real-world ophthalmic assessments. Prior automated glaucoma detection research has also emphasized their diagnostic utility [6, 23, 25], further supporting their inclusion. This expert-guided feature design ensures interpretability and clinical applicability of model outputs for automated glaucoma justification. The grading team was instructed to choose NRG regardless of co-morbidities such as age-related macular degeneration or diabetic retinopathy if they did not anticipate any glaucomatous optic neuropathy. Glaucoma severity was not recorded by the human graders because this study aimed to discover methods for automated screening.

Graders with sensitivity or specificity less than 80% or 95%, respectively, were excluded from this study. When a grader was excluded, all images they had labeled were reassessed by one of the remaining graders to maintain data integrity. Ultimately, 20 individual graders were included in the final group. For each eye, out of the three available CFPs, only the RG or NRG images were selected for inclusion in the dataset if they were available. To prevent data overlap between the training and test sets, the dataset was carefully divided into a training set comprising 101,442 CFPs and a test set containing 9,736 CFPs (See Table I). Comprehensive information about the acquisition process, labeling criteria, and dataset statistics, including the prevalence of different ethnic groups, can be found in [23]. The dataset represented a diverse population, including individuals of African descent, Asians, Whites, Native Americans, Latin Americans, people of mixed ethnicity, individuals from the Indian subcontinent, and those of unspecified ethnicity [23]. This diversity ensures that the dataset captures a wide range of demographic and clinical variability, contributing to its robustness and applicability in various contexts.

B. OHTS Dataset

The Ocular Hypertension Treatment Study (OHTS) [26] is a prospective, multicenter experiment (22 separate institutions in the United States) that aimed to postpone or prevent vision loss in ocular hypertension patients. The study obtained 66,742 fundus images (58,779 single-stereo and 7,963 pair-stereo) from 3272 eyes of 1636 participants over 16 years. At the baseline visit, all patients in the research appeared to have a normal optic disc and visual field. Previous works [27, 28] discussed the OHTS dataset and the glaucoma detection process in detail.

In this study, we used OHTS for external validation of the top three JustRAIGS participants' models. We selected the OHTS dataset because it provides a large, multi-center cohort, including cases that progressed from ocular hypertension to glaucoma over an extended follow-up period. Although OHTS primarily enrolled ocular hypertension cases, the subset we selected for validation explicitly includes confirmed glaucoma cases, rigorously diagnosed based on structural (optic disc deterioration) and functional (visual field) criteria at their final follow-up visits. This makes OHTS a particularly valuable external validation dataset for assessing whether our models generalize effectively across different disease severity stages, specifically capturing subtle optic disc changes indicative of glaucoma progression. Nevertheless, inherent distribution differences exist between JustRAIGS and OHTS datasets. These potential distribution shifts are explicitly summarized in Table II.

TABLE II
COMPARATIVE ANALYSIS BETWEEN JUSTRAIGS AND OHTS DATASETS.

Characteristic	JustRAIGS Dataset	OHTS Dataset
Study type	Retrospective, multi-center screening dataset	Prospective, multi-center clinical trial
Number of fundus images	111,178 images (full dataset); 9,736 images (test set)	66,742 images (full dataset); 5,000 images (test set)
Disease definitions	Referable glaucoma (RG) defined by expert evaluation of fundus images alone	Confirmed glaucoma diagnosed based on structural (optic disc) and functional (visual field) criteria
Dataset strengths	Large-scale, diverse image quality and population; focused on referral and justification of glaucoma	Longitudinal progression data, rigorous glaucoma confirmation with visual fields; suitable for assessing early glaucoma detection performance
Potential biases and distribution shifts	Bias toward structural definitions of glaucoma; lack of visual field data; diversity in image acquisition conditions (varied quality, artifacts)	Potential bias toward ocular hypertension cases; predominantly early-stage glaucoma; more homogeneous and standardized imaging conditions

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

We selected a subset that included 5000 CFPs of which 4,000 were from normal eyes and 1000 images were from eyes with glaucoma (due to optic disc or visual field abnormalities). They included three similar glaucomatous features with the JustRAIGS dataset. These features were *ThinningDiscRim_RT/LT*[5] and *NotchRim_RN/LN*[6] corresponded to "appearance neuroretinal rim superiorly" or "appearance neuroretinal rim inferiorly", *DiscHemorrhage-DH*[7] corresponded to "disc hemorrhage(s)", *LargeCup_LC*[8] corresponded to "large cup". We assumed that when one of or both *ThinningDiscRim_RT/LT* and *NotchRim_RN/LN* was 1, the label of "appearance neuroretinal rim superiorly" and "appearance neuroretinal rim inferiorly" as RG, and when both *ThinningDiscRim_RT/LT* and *NotchRim_RN/LN* were 0, we considered the label of "appearance neuroretinal rim superiorly" and "appearance neuroretinal rim inferiorly" as NRG. We used this annotation for doing task 2 on the OHTS dataset (See Table III).

TABLE III

ANNOTATION FOR DOING TASK 2 ON THE OHTS DATASET. APPEARANCE NEURORETINAL RIM SUPERIORLY = ANRS, APPEARANCE NEURORETINAL RIM INFERIORLY= ANRI, THINNINGDISCRIM_RT/LT= RT/LT, NOTCHRIM_RN/LN=RN/LN, LARGECUP_LC=LC, DISCHEMORRHAGE-DH=DH, DISC HEMORRHAGE(S)= DH(S), LARGE CUP=LC(S).

JustRAIGS				OHTS			
ANRS	**ANRI**	**LC(S)**	**DH(S)**	**RT/LT**	**RN/LN**	**LC**	**DH**
NRG	NRG			0	0		
RG	RG			0	1		
RG	RG			1	0		
RG	RG			1	1		
		NRG				0	
		RG				1	
			NRG				0
			RG				1

III. CHALLENGE SETUP

The JustRAIGS challenge comprises two distinct phases: development and testing. The development phase commenced on January 8th, 2024, and concluded on April 20th, 2024, allowing participants nearly three months to create their algorithms. At the beginning of this phase, the training set was made public and has been accessible for download on Zenodo[9] under the CC BY-NC-SA license. The test set remains closed, indicating that both the image data and labels are confidential and inaccessible for download. In the development phase, each team was permitted to make ten submissions, which accounted for 5% of the test set and were utilized for evaluation purposes. Additionally, all indicators related to the challenge were calculated and communicated to the participants. The test phase commenced on March 1, 2024, and concluded on April 20, 2024. In this phase, the complete test set was used to assess the performance of the submitted algorithms.

The algorithms were finally assessed by using the entire test dataset, and each team was limited to making just one submission (refer to Fig. 1).

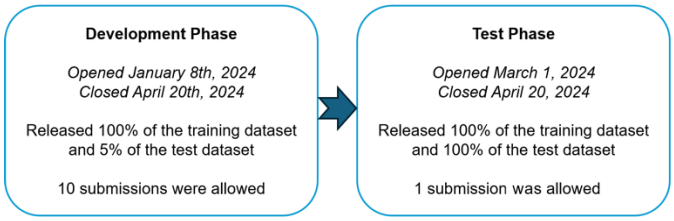


Fig. 1. An overview of the JustRAIGS challenge's phases.

The JustRAIGS competition had a total of 322 participants from 37 countries worldwide (refer to Fig. 2) in the development phase, and 20 teams successfully made it to the test phase. Participants were required to submit their trained algorithms in a Docker container to our challenge site for evaluation. This facilitates the execution of the submitted algorithms on test data that is not directly available to the participating teams. The code for creating a containerized submission is accessible on GitHub[10].

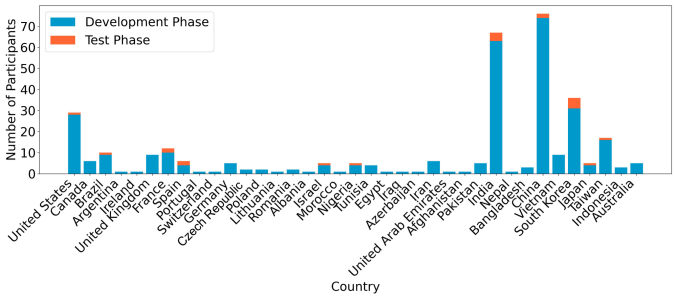


Fig. 2. An overview of the JustRAIGS challenge's participants. The bar chart shows the number of participants by country. The x-axis displays the country names, and the y-axis shows the number of participants. The number of participants in the development and test phases is shown in blue and orange colors, respectively.

[5] ThinningDiscRim_RT/LT is an annotation label used in the OHTS dataset to indicate that the optic nerve's neuroretinal rim has thinned in the right (RT) or left (LT) eye.

[6] NotchRim_RN/LN is an annotation label used in the OHTS dataset to denote the presence of a notch in the neuroretinal rim in the right (RN) or left (LN) eye.

[7] DiscHemorrhage-DH is an annotation label used in the OHTS dataset to denote the presence of a disc hemorrhage —a small, flame- or splinter-shaped bleed—typically located at the optic disc margin.

[8] LargeCup_LC is an annotation label used in the OHTS dataset to indicate the presence of a large optic cup; a structural feature strongly associated with glaucomatous optic neuropathy.

[9] https://zenodo.org/uploads/10035093

[10] https://github.com/DM2LL/JustRAIGS-IEEE-ISBI-2024

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

The algorithms were implemented on the challenge platform using an NVIDIA T4 GPU with 16 GB of VRAM and 8 CPUs with 32 GB of RAM. The algorithms were required to generate two outputs. One output was related to the performance of referral for glaucoma, specifically the continuous output indicating the likelihood of RG and NRG. The other output was related to justified glaucoma, specifically the multi-label classification of ten glaucomatous features.

A. Evaluation Metrics

We evaluated each algorithm on two distinct aspects: (1) referral performance (P_{ref}) and (2) justification performance (P_{just}). The evaluation of P_{ref} was conducted by measuring the sensitivity at a specificity level of 95% (SE@95SP). To determine the SE@95SP, first, the Receiver Operating Characteristic (ROC) curve is generated for each algorithm by plotting sensitivity versus (1 - specificity) across a range of threshold settings of the continuous output scores; then, the specific threshold is identified where the specificity is exactly 95% (i.e., the false positive rate is 5%) and calculates the corresponding sensitivity at this point, which reflects the algorithm's ability to correctly identify true positive cases (patients who should be referred for glaucoma) while maintaining a high level of specificity; this metric allows us to standardize the comparison of different algorithms by assessing their performance at a clinically relevant specificity level, ensuring that high sensitivity is achieved without a significant increase in false positives. To assess the effectiveness of P_{just}, we employed the Hamming Loss Average (HLA) metric, which calculates the proportion of mismatches between the predicted and the ground truth multi-label classification for ten additional features related to glaucoma. This metric was chosen because it effectively quantifies the performance of the models in identifying the correct combination of multiple glaucoma-related features, providing a comprehensive evaluation of their ability to generate accurate and clinically meaningful justifications for referral decisions. Formally, for N images and L features in the HLA metric, each model outputs a set of binary predictions $\hat{y}_{i,j}$ for the *jth* feature of the *ith* image, compared against a ground-truth label $y_{i,j}$. HLA measures the fraction of misclassifications across all instances and labels:

$$HLA = \frac{1}{N \times L} \sum_{i=1}^{N} \sum_{j=1}^{L} 1(\hat{y}_{i,j} \neq y_{ij}) \quad (1)$$

where $1(\cdot)$ is an indicator function that equals 1 if the predicted label differs from the truth and 0 otherwise. This straightforward metric penalizes every incorrect label, whether missing or extraneous, offering a holistic view of each model's ability to handle multiple features simultaneously.

To establish the final ranking, we initially assessed all participants based on the two metrics separately, P_{ref} and P_{just}, which led to the rankings R_{ref} and R_{just}, respectively. The final score, denoted as S_{final}, was computed by taking the average of the aforementioned ranks:

$$S_{final} = (R_{ref} + R_{just})/2. \quad (2)$$

The final ranking was determined by S_{final}, with a lower S_{final} value corresponding to a higher ranking.

IV. Participating Methods

Twenty teams advanced to the final test phase with their algorithms. Here, we briefly describe the strategies employed by each of the nine teams that made it to the top of the challenge leaderboard[11]. These nine teams were selected because they maintained clarity and conciseness while highlighting the most effective and diverse methodological approaches. The remaining 11 teams either did not surpass the minimum performance thresholds required for competitive analysis or did not complete the final containerized submission process. The ISBI challenge proceedings contained a selection of the participating methods [29-37]. Structured summaries of the participating approaches are provided in Table IV.

A. EVA2 [29]

The *EVA2* model proposed by the VUNO team featured a multi-stage region-based neural network framework comprising three primary models for segmentation, binary classification (task 1), and multi-label classification (task 2). The segmentation model extracted the optic disc region from fundus images using U-Net++ [38] with EfficientNet-B3 [39] as the backbone, ensuring that the classification models concentrate specifically on the optic disc area, which is crucial for glaucoma screening. The task 1 model classified images into referable and no referable glaucoma categories by independently training multiple classifiers with different backbones (EVA01 [40], SwinV2 [41], and ConvNext [42]) and employed a deep ensemble strategy to enhance generalization performance, with the models trained on cropped optic disc images. The task 2 model identified 10 specific glaucoma-related features using a custom loss function that incorporated a masked focal loss [43] and a recall term to address ambiguous class labels and class imbalance, with the ensemble technique applied to this task as well.

[11]https://justraigs.grand-challenge.org/evaluation/test-phase/leaderboard

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

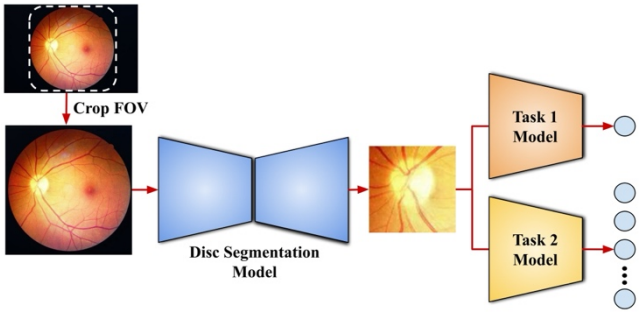


Fig. 3. Overview of the EVA2 model by the VUNO team. The pipeline uses a U-Net++ segmentation with an EfficientNet-B3 model to isolate the optic disc, followed by two classifiers: one for referable glaucoma detection (Task 1) using an ensemble of EVA01, SwinV2, and ConvNext models, and another for identifying ten glaucomatous features (Task 2). A custom loss function and post-processing step improve feature recall, making the model both accurate and interpretable.

For instances where the task 1 model predicted glaucoma, but the task 2 model predicted all features as normal, the highest probability feature prediction of task 2 was forcibly set to abnormal to improve the model's recall and ensure it captures potential glaucoma features. The task2 model is applied to the task-arithmetic technique to boost the performance of 10 multi-label tasks. An overview of the EVA2 model is shown in Fig. 3.

B. DC-LS [30]

As computing capabilities advance, modern machine learning and computer vision systems often grow more complex with larger models and sophisticated optimization strategies. However, significant improvements can also be made by focusing on better understanding and utilizing existing training data and annotations. This data-centric approach can result in substantial performance gains, sometimes exceeding those achieved by merely increasing model complexity. The *DC-LS* model uses this strategy.

The *DC-LS* architecture for glaucoma screening from eye fundus images utilized a ResNet50 [44] model as the baseline framework for both tasks of this challenge. This model was optimized with an Adam algorithm, by using a default learning rate of 1×10–4 and a batch size of 8, continuing training until overfitting was detected on a validation set consisting of 20% of the data. The architecture aimed to balance complexity and accuracy. A data-centric label smoothing approach was employed to encode inter-rater disagreement by adjusting the labels according to the agreement and expertise of the annotators. The model was trained on preprocessed data with label smoothing, with the dataset divided into 80% for training and 20% for validation. Additionally, a 5-fold training ensemble was implemented to enhance the model's robustness. An overview of the DC-LS model is shown in Fig. 4.

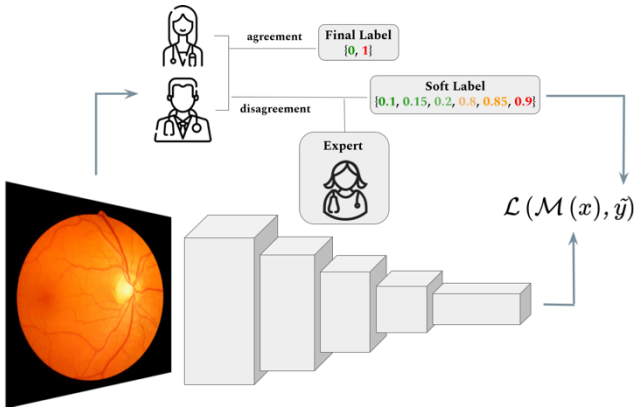


Fig. 4. Overview of the DC-LS model using a ResNet50 backbone. The approach emphasizes data-centric learning by incorporating label smoothing to reflect annotator disagreement and expertise. Labels are softened when graders disagree and finalized when a consensus is reached. This strategy, combined with a 5-fold ensemble and optimized training, enhances model robustness and improves generalization without increasing model complexity.

C. Danp [31]

The *Danp* model utilized several deep learning models, including ConvNeXt-Tiny [42], ConvNeXt-V2-Tiny [45], EVA02-Base [46], and EfficientNet-B7 [39], which were trained with large input image sizes (896 and 1024 pixels) rather than focusing solely on the optic disc region. The model training techniques involved combining the binary target for referable glaucoma with 10 additional glaucoma-related features, resulting in an 11-element target vector composed of binary values. To optimize the model, each element in the target vector was treated as an independent binary classification task, utilizing binary cross-entropy loss. Positive class samples were assigned higher weights to ensure balanced sampling, making them more frequently represented during training. A series of image augmentations, such as CLAHE [47], horizontal flip, image rotation, sharpening filter, and contrast adjustment, were employed to enhance robustness. To enhance the performance of the models, the authors implemented test time augmentation (TTA) [48]. The dataset was divided into 5 stratified folds based on the referable glaucoma target variable, by using patient ID as the grouping parameter to maintain consistency. During inference, the best-performing models from different training folds were combined into an ensemble using a hard voting strategy. In cases of grader disagreement, the soft labeling transformation averaged the label across graders; for instance, if G1 = RG, G2 = NRG, and G3 = NRG, the soft consensus label is 0.333. Soft labeling transformation was applied to both the referable glaucoma label and 10 additional features.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

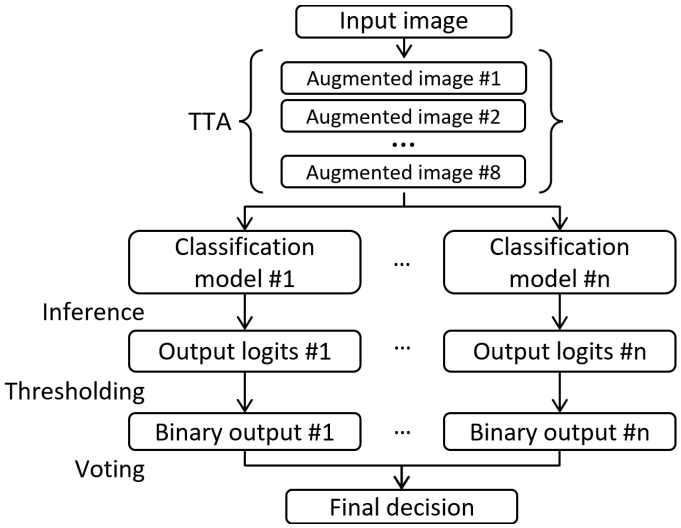


Fig. 5. Block diagram of the Danp method. The model employs Test-Time Augmentation (TTA) to generate multiple image variants, which are then processed by an ensemble of deep learning models, including ConvNeXt-Tiny and EVA02-Base. Each model's predictions are thresholded into binary outputs and aggregated via majority voting. This strategy improves robustness and accuracy in referable glaucoma and feature detection by leveraging ensemble learning and soft labeling.

After model training, the best model for each fold was selected based on its cross-validation ranking, which was calculated by using an evaluation metric aligned with the challenge's leaderboard ranking method. The ConvNeXt-Tiny [42] architecture consistently achieved the highest rankings across all folds. A block diagram of the Danp method is shown in Fig. 5.

D. Segnet [32]

The *Segnet* architecture employed a deep learning approach with two CNN models. The first model was a binary classification model based on the EfficientNet-B4 architecture. It classified the input cropped optic disc image to determine the presence of RG or NRG. This referral model was trained by using binary cross-entropy loss. The second model was a multi-label classification model based on the DenseNet201 architecture. It identified the presence of 10 glaucomatous features in the input images labeled as referable glaucoma. This justification model was trained only with positive samples since it was invoked only for images labeled as referable glaucoma. To tackle the issue of class imbalance, they utilized data sampling and domain adaptation strategies. The data was split into 80% for training and 20% for validation. The training subset was balanced by resampling and shuffling negative samples after every epoch to expose the model to all data. Additionally, a disc segmentation model [25] was used to segment the optic disc region from the input fundus images, focusing the subsequent models on this critical area for glaucoma detection. The final ground truth labels for the justification task were derived in the following manner: if both specialists G1 and G2 reached a consensus on a feature, the feature was assigned the mutually agreed-upon value. If there was a disagreement between G1 and G2 over a feature value, and specialist G3 had provided a label for the feature, the feature would be assigned based on G3's label. If G1 and G2 had conflicting opinions on a feature and there was no label from G3, the feature was designated as NaN (not a number). The NaN values were excluded or hidden while calculating the loss and metrics.

E. Brighteye [33]

The proposed method, *Brighteye*, for glaucoma screening with color fundus photographs included several key components. YOLOv8 was used to detect and locate the optic disc in the fundus images, utilizing the predicted center coordinates, width, and height to crop the region of interest (ROI) around the optic disc. Once detected, the ROI around the disc center was cropped, and the background was removed to focus on the relevant region. The dataset was divided into training and validation sets at a 4:1 ratio, incorporating all RG cases and 4000 randomly selected NRG cases. The training loss was calculated as the average of two binary cross-entropy losses from the model outputs. Data augmentation techniques, including horizontal and vertical flips, rotations, and adjustments to saturation, brightness, and hue, were applied to enhance generalizability and prevent overfitting. A Vision Transformer (ViT) based model [49] was employed for glaucoma detection, dividing the input image into patches, projecting them into a 1D vector, and using self-attention mechanisms to learn long-range relationships among pixels. Additionally, ten binary classifiers, sharing the same architecture as the glaucoma detection model, were used to independently predict the presence of specific glaucomatous features.

F. JustViT [34]

The *JustViT* method for automated detection of glaucoma and diagnostic features in the JustRAIGS challenge involved a deep learning pipeline with several key components. YOLOv8 was employed for optic disc segmentation, followed by region of interest (ROI) cropping. Initially, it was trained on 1,000 masks generated by a U-Net[12]. After this first round of training, YOLOv8 was used to generate approximately 90,000 masks from the remaining images. The model was then retrained on the full set of masks, achieving a segmentation success rate of approximately 99.6%. The images were trimmed of excessive black space, padded to

[12] https://github.com/Paresh-shahare/Optic-Disc-Segmentation

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

maintain a 1:1 aspect ratio, resized to 2,000*2,000 pixels, and enhanced by using Contrast Limited Adaptive Histogram Equalization (CLAHE) [50]. Next, the images were augmented with Auto Augment [51]. The dataset was divided into training and validation sets, with the training set for glaucoma detection including 100,769 images (2,943 RG and 97,827 NRG) and the validation set comprising 654 images (327 RG and 327 NRG). For feature detection, the subset with RG labels (3,270 images) was split into training (2,935 images) and validation sets (327 images). If the initial two graders disagreed on any additional features and the third grader's annotation was not available, the most frequently observed label for the contested feature within the dataset was used. Four Vision Transformers (ViTs) [49], pre-trained on ImageNet and fine-tuned on different subsets of the dataset, were employed for glaucoma detection and diagnostic feature classification. These ViTs used a patch size of 16 and shared the same hyperparameters, including weighted cross-entropy loss with class inverse distribution weights, a learning rate of 1.0e-4 for the network, and a learning rate of 1.0e-5 for the head.

G. UR-UdC [35]

The *UR-UdC* model proposed an ensemble technique containing ConvNext and Swin models for the binary task and ConvNext for the multi-label task. These models were pre-trained on ImageNet, and their head layers were replaced and retrained to classify fundus images. Training occurred in two stages: first, the new head was trained with the rest of the model layers frozen, and second, the entire model was retrained with all layers unfrozen. The difference between multi-label classification with binary classification was that the model's head contained 10 outputs, each corresponding to a specific glaucomatous characteristic. The multi-label classification task employed a loss function of binary cross-entropy with logits loss, which involved the combination of a sigmoid layer and binary cross-entropy loss. The SegFormer model [52] was used to extract masks for the optic disc and cup[13]. The dataset was split into 80% for training and 20% for validation, with subsampling used to balance the highly imbalanced dataset of 98,153 NRG and 3,270 RG cases. Data augmentation techniques, such as vertical and horizontal flips, rotations, zooms, and lighting transformations, were applied during training. To enhance the performance of the models, the authors implemented test time augmentation (TTA) [48]. This involved generating five random variations of an image, such as vertical and horizontal flips, rotations ranging from -180º to 180º, zooms, and lighting transformations. They then made predictions on these modified images using a model and calculated the average of these predictions.

H. LaTIM [36]

LaTIM was another model proposed in the JustRAIGS challenge. In the data preprocessing stage of this model, black borders were removed from images, which were then resized, and their intensity distribution was normalized to enhance neural network processing. Data augmentation techniques included random horizontal and vertical flips, rotations, and color adjustments in brightness, contrast, and hue. For the binary classification task, the model architecture was an ensemble of five ResNet50 [53] models, where each model was independently trained with the same structure, starting without freezing any layers and using the AdamW optimizer with an initial learning rate of 0.001, dynamically adjusted during training. Each one was trained on unique data subsets, and the final predictions were averaged across the five models' outputs to leverage diverse decision patterns, boosting robustness and performance. For the multi-label classification of ten additional glaucomatous features, an ensemble of the Eva Large Transformer model [40], Deit3 Transformer model [54], and ResNet50 CNN [53] with added multi-label classification heads were utilized. These models captured fine-grained, local features, complex dependencies between image regions, and global structural features, respectively. Training employed binary cross-entropy loss with the AdamW optimizer at different learning rates: 1e-3 for ResNet50, 1e-4 for Deit3, and 1e-5 for Eva. Predictions from the three models were averaged to integrate their diverse strengths.

The ResNet50 ensemble that was suggested was compared to various baseline models, including a fine-tuned RetFound model [8]. The RetFound model is a foundational model that has been pre-trained on more than 1.6 billion images. The ResNet50 ensemble model demonstrated greater performance, especially when presented with higher-resolution images, despite its comprehensive training. In Task 1, the ResNet50 ensemble model outperformed other models due to its higher resolution (e.g., 800×800 vs. 224×224). This shows that image resolution is vital for improving the model's sensitivity, especially for capturing fiber optic nerve head characteristics that may be necessary to distinguish healthy from glaucomatous eyes. This supports prior studies showing the usefulness of high-resolution images in medical image analysis jobs where minor differences are diagnostic [55].

[13]https://huggingface.co/pamixsun/segformer_for_optic_disc_cup_segmentation

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

TABLE IV

OVERVIEW OF PARTICIPANTS' METHODS.

Algorithm	Pre-processing	Model architecture	Handling strategy for ambiguous class labels in multi-label classification task
EVA2	• Crop field of view (FOV) • Optic disc segmentation with U-Net++ encoded with EfficientNet-B3 • Data augmentation	• Deep ensemble strategy with backbones EVA01, SwinV2, and ConvNeXt • Combination with multi-task models with task-arithmetic technique	• A custom loss function included a masked focal loss and a recall term • Correct the output of task1 model when task2 model predicts positive at least one class
DC-LS	• Crop field of view (FOV) & resize to 512x512 • Data augmentation: Flips, Scaling, Rotation, Brightness/Color Perturbations	• ResNet50 architecture is optimized by minimizing CE using an Adam algorithm, cyclical learning rate 1e-4, 15 epochs, early stopping based on Average Precision	• Data-centric label smoothing approach
Danp	• Balanced weighted sampling • Data augmentation	• Deep ensemble strategy with backbones ConvNeXt-Tiny, ConvNeXt-V2-Tiny, EVA02-Base, and EfficientNet-B7	• Soft labeling transformation
Segnet	• Optic disc segmentation utilizing a pretrained model • Data sampling and domain adaptation strategies to address the class imbalance • Data augmentation	• EfficientNet B4 architecture with cross-entropy loss for referral task • DenseNet201 architecture with cross-entropy loss for justification task, previously trained on the referral task	• Higher expert's vote labeling • Conflicting labels without a higher expert vote were set as NaN (not a number) and excluded from loss and metrics calculations
Brighteye	• Crop field of view (FOV) • Optic disc segmentation with YOLOv8 • Data augmentation	• Vision Transformer (ViT)	• Ten binary classifiers were used to predict the presence of specific glaucomatous features, independently.
JustViT	• Crop field of view (FOV) • Resize and CLAHE contrast enhancement • Optic disc segmentation with YOLOv8 • Crop region of interest (ROI)	• Four Vision Transformers (ViTs) • YOLOv8	• If third grader's annotation was not available, the most frequently observed label for the contested feature within the dataset was used.
UR-UdC	• Optic disc segmentation with SegFormer	• Ensemble strategy with backbones ConvNeXt and Swin for referral task • ConvNeXt architecture for justification task	-
LaTIM	• Crop field of view (FOV) • Data augmentation	• Ensemble of five ResNet50 models for referral task • Ensemble of Eva, Deit3, and ResNet50 for justification task	• Prioritizing the third grader's (G3). If the G3 label was missing, the maximum value between the first (G1) and second (G2) graders' labels was selected.
Roshni	• Data augmentation • Optic disc segmentation with YOLOv8	• Swin Transformer Model with cross-entropy loss	-

In Task 2, by using many models in an ensemble considerably minimizes the Hamming loss, resulting in more accurate categorization. By using distinct model architectures (e.g., Eva, Deit3, ResNet50) and processing photos at various resolutions (e.g., 336x336, 384x384, 800x800), one can capture glaucoma-related ocular feature fluctuations. The ensemble method combines model strengths to mitigate shortcomings and create a reliable diagnostic tool. Ensemble learning improves medical image classification accuracy and resilience [56].

I. Roshni [37]

The Roshni model utilized the Swin Transformer Model (STM) [57] and was initially pre-trained on the COCO and ImageNet datasets and then fine-tuned on the REFUGE challenge dataset for glaucoma classification. This method avoids the requirement of pre-processing input photos to a consistent size. The method architecture involved YOLO V8 for optic disc detection. For Task 1, the model used a cross-entropy loss function with weights of 0.87 for RG and 2.5 for NRG, optimized with Adam over 15 epochs. Task 2 employed the BCEWithLogitsLoss function, optimized with Adam over 15 epochs, with an output threshold of 0.4 by using the sigmoid function. Data augmentation techniques, such as random horizontal flips, rotations (0, 180 degrees), affine transformations, and color jitter, were applied to enhance model robustness and adapted to real-world imaging variations. The model's performance was evaluated by predicting the presence of ten specific glaucomatous features in images labeled as referable glaucoma.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

V. Results

This section describes the glaucoma referral and justification performances of the nine participating teams in the final test phase. The final rankings, Sfinal, SE@95SP (Position), and HLA (Position) of the teams are shown in Table V.

TABLE V

TEST PHASE RESULTS ON THE CHALLENGE LEADERBOARD.

Algorithm	#Rank	S_{final}	SE@95SP (Position)	HLA (Position)
EVA2	1	2	0.8430 (3)	0.125 (1)
DC-LS	2	3	0.8450 (2)	0.1410 (4)
Danp	3	3.5	0.8500 (1)	0.1480 (6)
Segnet	4	6	0.7970 (10)	0.1370 (2)
Brighteye	5	6.5	0.7850 (11)	0.1370 (2)
JustViT	6	7	0.7980 (8)	0.1480 (6)
UR-UdC	7	8	0.8340 (5)	0.2030 (11)
LaTIM	8	9	0.8180 (6)	0.2540 (12)
Roshni	9	10	0.8180 (6)	0.2560 (14)

A. Glaucoma Referral Performance

The top three teams with the highest SE@95SP scored expert-level referral performances on the JustRAIGS test set with a sensitivity of 0.84 (95% CI: 0.83 – 0.85), 0.85 (95% CI: 0.83 – 0.86), and 0.85 (95% CI: 0.84 – 0.86) at 95% specificity by EVA2, DC-LS, and Danp, respectively. Confusion matrices of the top three teams are illustrated in Fig. 6.

The glaucoma referral performances of the participating teams are summarized in Table V by SE@95SP. The highest SE@95SP was 0.8500. This score was achieved by the *Danp* model.

B. Glaucoma Justification Performance

The best HLAs that were achieved by any of the top three teams were 0.13 (95% CI: 0.12 – 0.13), 0.14 (95% CI: 0.13 – 0.15), and 0.15 (95% CI: 0.14 – 0.16), by EVA2, DC-LS, and Danp, respectively. Box Plots of the HLA for the top three teams are illustrated in Fig. 7.

The justification performances of the participating teams are summarized in Table V, showing the HLA. The best score for HLA was 0.1250. This score was achieved by the *EVA2* model, which was proposed by the VUNO team.

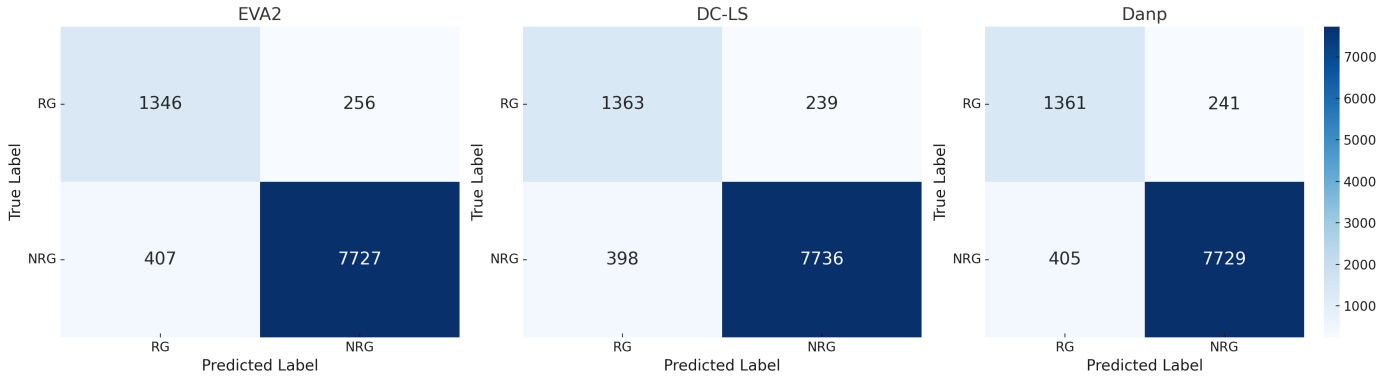


Fig. 6. Confusion matrices of the top three teams with the highest SE@95SP score expert-level referral performances on the JustRAIGS test set. Confusion matrices in the first, second, and third columns indicate the sensitivity of 0.84 (95% CI: 0.83 – 0.85), 0.85 (95% CI: 0.83 – 0.84), and 0.85 (95% CI: 0.84 – 0.86) at 95% specificity for EVA2, DC-LS, and Danp algorithm, respectively.

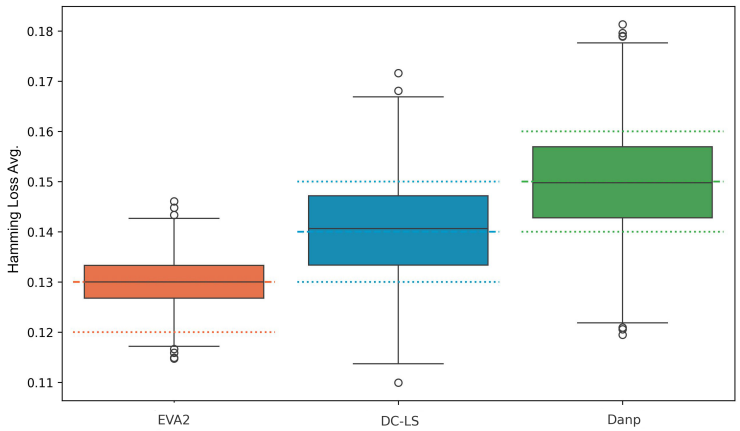


Fig. 7. Box Plots of HLA for the top three teams score expert-level justification performances on the JustRAIGS test set. Box Plots from left to right indicate a Hamming loss average of 0.13 (95% CI: 0.12 – 0.13), 0.14 (95% CI: 0.13 – 0.15), and 0.15 (95% CI: 0.14 – 0.16) for EVA2, DC-LS, and Danp algorithms, respectively.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

To gain deeper insight into justification quality, we computed per-feature precision, recall, and F1-scores for the top three models (EVA2, DC-LS, and Danp). Table VI summarizes the results across the ten glaucoma-related features. Across all three metrics, EVA2 consistently achieved the strongest performance, particularly excelling in detecting large cups (Precision/Recall/F1-score: 0.74/0.71/0.72), disc hemorrhages (0.70/0.67/0.68), and nasalization of the vessel trunk (0.68/0.65/0.67). While DC-LS and Danp achieved competitive results, they exhibited slightly lower precision, recall, and F1-scores, especially for neuroretinal rim and nerve fiber layer defect detection.

TABLE VI

PER-FEATURE PERFORMANCE (PRECISION / RECALL / F1-SCORE) OF GLAUCOMA-RELATED FEATURES FOR THE TOP THREE MODELS.

Features	**EVA2** Precision / Recall / F1-Score	**DC-LS** Precision / Recall / F1-Score	**Danp** Precision / Recall / F1-Score
Appearance Neuroretinal Rim Superiorly	0.64 / 0.61 / 0.63	0.63 / 0.60 / 0.61	0.62 / 0.58 / 0.60
Appearance Neuroretinal Rim Inferiorly	0.63 / 0.60 / 0.61	0.61 / 0.59 / 0.60	0.60 / 0.57 / 0.58
Baring of Circumlinear Vessel Superiorly	0.58 / 0.55 / 0.56	0.56 / 0.53 / 0.54	0.55 / 0.52 / 0.53
Baring of Circumlinear Vessel Inferiorly	0.59 / 0.56 / 0.58	0.57 / 0.54 / 0.55	0.56 / 0.53 / 0.54
Disc Hemorrhages	0.70 / 0.67 / 0.68	0.69 / 0.66 / 0.67	0.68 / 0.65 / 0.66
RNFL Defect Superiorly	0.66 / 0.63 / 0.65	0.64 / 0.62 / 0.63	0.63 / 0.60 / 0.61
RNFL Defect Inferiorly	0.64 / 0.61 / 0.63	0.62 / 0.60 / 0.61	0.61 / 0.58 / 0.59
Nasalization of Vessel Trunk	0.68 / 0.65 / 0.67	0.67 / 0.64 / 0.65	0.66 / 0.63 / 0.64
Laminar Dots	0.62 / 0.59 / 0.60	0.61 / 0.57 / 0.59	0.59 / 0.55 / 0.57
Large Cup	0.74 / 0.71 / 0.72	0.72 / 0.69 / 0.70	0.70 / 0.67 / 0.68

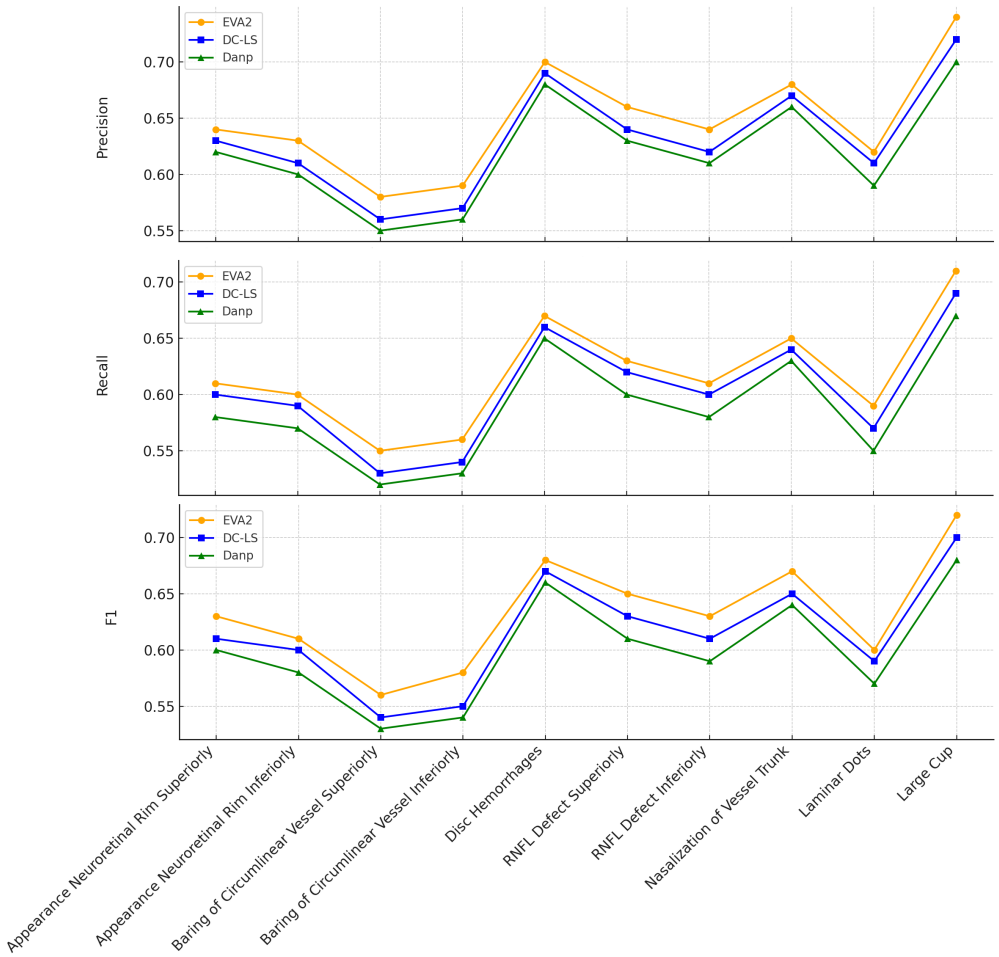


Fig. 8. Per-Feature Precision, Recall, and F1-Score comparisons of the ten glaucoma-related features across the three highest-ranking models (EVA2, DC-LS, and Danp). EVA2 consistently achieves superior per-feature performance, especially in critical features such as disc hemorrhages, nasalization of vessel trunk, and large cup detection. The trends reveal subtle differences in feature-wise justification quality, emphasizing EVA2's more consistent and clinically robust multi-label predictions compared to DC-LS and Danp.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

These findings highlight that EVA2 provided more accurate and clinically reliable multi-label justifications across diverse glaucomatous signs, reinforcing its top ranking in both referral and justification performance tasks. Fig. 8 illustrates per-feature Precision, Recall, and F1-Score comparisons of the ten glaucoma-related features across the three highest-ranking models.

C. Statistical Significance of Model Comparisons

To assess the robustness of performance differences among the top-performing models, we conducted statistical significance testing on both the referral and justification tasks. Specifically, we compared SE@95SP and HLA across EVA2, DC-LS, and Danp using 250 bootstrap samples and paired t-tests. Table VII reports the resulting p-values for each model pair. All comparisons revealed statistically significant differences ($p < 0.05$), confirming the reliability of the leaderboard rankings. Notably, the SE@95SP difference between EVA2 and Danp was highly significant ($p \approx 0.0029$), as was the HLA difference between DC-LS and Danp ($p \approx 0.0065$). These results demonstrate that observed model performance gaps, particularly in multi-label justification accuracy, are robust and not attributable to chance, reinforcing confidence in the validity of our evaluation.

TABLE VII

P-VALUES FROM STATISTICAL COMPARISONS (250 BOOTSTRAP SAMPLES).

Comparison	SE@95SP (p-value)	HLA (p-value)
EVA2 vs DC-LS	0.0379	0.0082
EVA2 vs Danp	0.0029	0.0490
DC-LS vs Danp	0.0497	0.0065

D. Computational Complexity Analysis

To further contextualize the practical deployment considerations of the three top-ranking methods from the JustRAIGS challenge, we conducted a computational complexity analysis. Table VIII presents a comparison of the three best-performing methods regarding their approximate number of parameters, inference time per image, and GPU memory requirements measured on an NVIDIA T4 GPU.

The DC-LS model, based solely on ResNet-50 architectures, was notably more efficient in terms of inference speed and memory consumption, reflecting the practical advantages of simpler CNN-based ensembles. EVA2 and Danp leveraged more complex, transformer-based architectures and larger CNN backbones, while computationally more demanding, but showed competitive or slightly superior performance. This comparison underscores the importance of balancing computational efficiency with performance accuracy, especially in clinical environments with limited computational resources.

TABLE VIII

COMPUTATIONAL COMPLEXITY ANALYSIS FOR THE TOP THREE MODELS.

Algorithm	Model Architecture	Number of Parameters (Millions)	Inference Time (ms/image)	GPU Memory Usage (GB)
EVA2	Ensemble of EVA01, SwinV2, ConvNeXt, U-Net++ (EfficientNet-B3)	~278	320 ± 15	8.5 ± 0.4
DC-LS	ResNet-50 (5-fold ensemble)	~125	100 ± 10	4.2 ± 0.3
Danp	Ensemble of ConvNeXt-Tiny, ConvNeXt-V2-Tiny, EVA02-Base, EfficientNet-B7	~208	280 ± 20	7.8 ± 0.5

E. Additional Evaluation on the External Data Set (OHTS)

Based on our external evaluation of the models using the OHTS dataset, the top three algorithms achieved a sensitivity of 0.87, 0.87, and 0.88 at 95% specificity, respectively. Moreover, the HLA of the top three algorithms were 0.11, 0.12, and 0.12, respectively.

VI. DISCUSSION

While some research has suggested that AI models are highly promising in diagnosing glaucoma in CFPs, the majority of these studies have not used real-world data to back up their claims. To achieve this objective, we depended on the community to create justified referral AI solutions for glaucoma screening. These solutions were developed by using the largest multi-center real-world CFP dataset that contains glaucoma labels. The JustRAIGS challenge was structured around this dataset to ensure that the generated algorithms can be easily utilized in a cloud-based context. The JustRAIGS challenge consisted of justified referral tasks for glaucoma in two stages: development and test phases. A total of 322 teams participated in the development phase, and out of those, 20 teams successfully advanced to the test phase. We utilized the top three techniques to evaluate their generalization on another extensive dataset.

A. Overall Findings

Eight of nine teams achieved a sensitivity of at least 80% and a specificity of at least 95%, which was the minimal performance required by the human graders who were regularly monitored during the grading process. This demonstrates that these models may achieve comparable performance to human graders in glaucoma screening, indicating that AI has the potential to be utilized in an automated screening process.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

Additionally, we assessed the referral efficacy of the top three algorithms on an external test dataset. Despite being trained on JustRAIGS data, the algorithms demonstrated exceptional performance on an external test dataset, indicating consistent and replicable results across diverse sets and populations. The results were significantly better on the OHTS dataset than on the JustRAIGS test set. This comprehensive generalization of the established solutions also demonstrates the capacity of algorithms trained on our dataset to be effectively applied in screening programs without any significant decrease in performance. The JustRAIGS dataset differs from the OHTS dataset in that it specifically represents a screening population. The OHTS population represents individuals at higher risk for developing primary open-angle glaucoma due to ocular hypertension, making the dataset particularly relevant for evaluating glaucoma screening algorithms in a cohort that is asymptomatic yet at increased risk. The JustRAIGS dataset probably has a higher proportion of patients with less severe glaucoma compared to clinical populations. The increased performance on the OHTS dataset compared to the JustRAIGS test dataset may be attributed to differences in the characteristics of the two datasets—factors such as variations in image quality, population risk profiles, and the presence of ocular features associated with ocular hypertension could make glaucoma detection more straightforward in the OHTS dataset. Since participants in the OHTS study have elevated intraocular pressure, they may exhibit subtle optic nerve head changes or other signs that the algorithms can detect more readily, even if they do not have advanced glaucoma.

On the other hand, although OHTS only labels three features that overlap with our ten-item multi-label classification scheme, we restricted the multi-label assessment to those three features (e.g., appearance neuroretinal rim superiorly or appearance neuroretinal rim inferiorly, disc hemorrhages, large cup) and treated the remaining seven labels as “not assessable” within OHTS. Despite this limitation, the partial alignment still reveals how well each model generalizes to at least the shared disc-related attributes. While not a perfect 1-to-1 mapping, it nonetheless provides practical insights into cross-dataset performance and underscores the potential utility of these models in clinical contexts where labeled features vary.

Each team's justification of ten glaucomatous features in the JustRAIGS test dataset was assessed by means of the HLA metric. The HLA metric measures the fraction of incorrectly predicted labels to the total number of labels in multi-label classification tasks, providing a comprehensive assessment of a model's performance; in our study, we used HLA to evaluate each team's ability to accurately identify the ten glaucomatous features that justify referral decisions, and the low HLA scores achieved by the top-performing algorithms on both the JustRAIGS test dataset and the external OHTS dataset are encouraging, indicating that the models may make errors in predicting the presence or absence of these features. This strong performance suggests that the algorithms are effective not only in detecting glaucoma but also in providing accurate and clinically meaningful justifications, which is highly useful in a screening context where precise identification of glaucomatous features enhances interpretability, builds clinician trust, and improves patient outcomes by facilitating early and appropriate interventions.

We also assessed the justification performance of the top three performing algorithms on the external OHTS dataset using the HLA metric, which measures the fraction of incorrectly predicted labels to the total number of labels in multi-label classification tasks despite being trained solely on the JustRAIGS dataset, the algorithms maintained low HLA scores on the OHTS dataset, indicating that they accurately identified the ten glaucomatous features associated with each image even in a different population. These results on an external dataset demonstrate strong generalization for the justification task, suggesting that the models can reliably provide accurate and clinically meaningful explanations for their glaucoma referral decisions across diverse datasets, which is essential for their utility in real-world screening programs.

There is a significant disparity in performance among the teams participating in both the referral and justification tasks. Hence, it is crucial to ascertain the primary methodological decisions made by the teams that achieved the highest performance. A prominent distinction between the top three participant teams and the remaining contestants was their advanced architectures or data-centric strategies. Transformers may outperform CNNs due to their effective modeling of long-range dependencies [58, 59], facilitating a better global comprehension of contextual information in medical images [59]. Specifically, for glaucoma detection, transformers can more effectively capture the global structural patterns of the optic nerve head and retinal nerve fiber layer in fundus photographs, which are critical features for identifying glaucomatous damage [60]. Indeed, literature reports that transformer-based or hybrid Transformer–CNN methods often surpass conventional CNNs alone in medical image analysis [61-63]. However, our findings also highlight that a carefully optimized CNN approach, as demonstrated by the DC-LS team, can still be highly competitive. By combining a data-centric label smoothing strategy with thorough data preprocessing, DC-LS achieved performance on par with the transformer-based teams, underscoring that effective data

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

handling and model tuning can compensate for, or even rival, purely architectural advantages.

Furthermore, some participants cropped the FOV around the OD in the pre-processing step. This appears to be a viable approach for attaining improved screening efficacy. One possible explanation for the efficacy of this strategy is that the majority of glaucoma-related characteristics are located on or around the OD.

Our study also revealed that incorporating ensemble learning methods improved performance across all metrics up to a specific limit. Beyond this point, adding more models to the ensemble resulted in decreased performance. This decline may be due to the fact that the additional models performed poorly enough to diminish the ensemble's overall effectiveness instead of enhancing it. Alternatively, this could indicate that the method used to combine the outputs of the underlying models is suboptimal. Generally, ensemble learning works by effectively combining multiple weak models into a stronger one; however, if the combination strategy is flawed, adding more models, even if they are weak learners, may not enhance and could even degrade performance. This suggests that refining the ensemble aggregation method could potentially mitigate the observed performance deterioration when incorporating additional models.

Finally, the data-centric label smoothing approach showed benefits by improving model generalization and robustness. This approach enhanced traditional label smoothing by considering the inherent data distribution and uncertainties rather than applying a uniform adjustment across all samples. It helped mitigate overconfidence in predictions, particularly for our imbalanced dataset, by reducing the impact of hard-label errors. As a result, it reduced overfitting and could better capture nuanced patterns in the data, leading to improved performance on test data. Additionally, it could enhance the interpretability and reliability of the model.

B. Detailed Error Analysis

To investigate which image characteristics consistently led to classification errors, we closely examined the confusion matrices (Fig. 6) for EVA2, DC-LS, and Danp.

For false positives (FP), we found that many poor-quality images exhibited underexposure, obscured fundus details due to media opacities, or general low contrast. Such factors can mimic signs of glaucomatous disc changes (e.g., a faint neuroretinal rim), prompting the models to issue false alarms. Furthermore, some images had borderline disc cupping or subtle rim thinning. Although these eyes were determined by expert graders to be non-glaucomatous, they were frequently misclassified by the algorithms.

For false negatives (FN), we noted a similar pattern, like images showing very early or mild optic nerve head changes (e.g., slight rim thinning, minimal nerve fiber layer defects) could be incorrectly labeled as negatives. This concurs with the longstanding challenge of detecting early glaucoma solely from color fundus images.

C. Strengths and Limitations

This study presents several strengths. First, it introduces a dual-task benchmarking framework, combining binary referral prediction with multi-label justification, that closely mirrors real-world clinical workflows. This dual-task design encourages the development of interpretable AI systems and advances beyond previous glaucoma screening challenges that focused solely on binary classification. Second, the challenge uses a large, high-quality dataset, curated and annotated by expert graders, and evaluated using clinically meaningful metrics such as SE@95SP and HLA. Finally, the inclusion of an external validation set (OHTS) helps evaluate generalization performance across populations.

However, there are also limitations. One notable limitation is that referable glaucoma labels were derived solely from fundus photographs, without confirmation from functional assessments such as visual field testing. As this study is designed for screening purposes rather than definitive diagnosis, early-stage glaucoma cases without clear structural damage may have been labeled as non-referable. Additionally, the ten glaucomatous features used for justification were manually annotated, and while the graders were well-trained, the process may have introduced label variability due to inter-rater differences or subjective interpretation of subtle signs.

Another important limitation is the potential presence of co-existing ocular conditions in the dataset, such as myopia, diabetic retinopathy, or non-glaucomatous optic neuropathies. The dataset was sourced from a population-based diabetic retinopathy screening program across over 500 sites in the United States, without exclusion criteria for these comorbidities. Graders were not instructed to remove or label such cases separately. These co-pathologies may mimic or obscure glaucomatous changes and can contribute to algorithmic false positives or false negatives. While the reported SE@95SP metric inherently captures the influence of these confounding factors on model performance, future work may benefit from integrating multimodal data (e.g., OCT, visual fields) to improve specificity and overall clinical reliability.

Despite these limitations, the challenge offers a meaningful step forward in the development of trustworthy, interpretable, and clinically relevant AI models for glaucoma screening.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

D. Future Directions

Considering the proposed solutions from the teams, we believe it is worthwhile to continue developing the ideas put up by the participants. As previously indicated, the EVA2 algorithm gained the best HLA score among all participants. The DC-LS algorithm achieved a high score in terms of accurately diagnosing glaucoma utilizing domain expertise and effective management of labels for this specific part of the task. Furthermore, the Danp algorithm demonstrated a remarkably efficient approach for diagnosing glaucoma through referrals, as evidenced by their SE@95SPs being significantly higher than those of the other contestants.

Although the overall justification performance, as measured by HLA, was strong, we acknowledge that the precision and recall for certain individual features remain modest, particularly for subtle structural signs such as rim defects or laminar dots. These values may currently fall short of clinical expectations for feature-level diagnostic reliability. This limitation reflects inherent subjectivity in feature annotation and the challenges of modeling fine-grained variations in fundus images. To improve performance on these clinically relevant features, future work could explore several strategies. First, label quality could be enhanced through consensus-based annotation or confidence-weighted labels to reduce inter-grader variability. Second, feature-aware training objectives such as focal loss or feature-specific weighting could be used to emphasize underperforming labels. Additionally, multimodal integration (e.g., combining CFPs with OCT or visual field data) may help models better link subtle structural features with functional deficits. Progressive training strategies may also help models gradually learn harder-to-classify features by prioritizing easier ones early in training. These directions may advance the feature-level reliability required for real-world deployment in clinical workflows.

In addition to determining the existence of RG and NRG, the graders were requested to indicate the specific clinical signs of glaucoma observed in the eyes categorized as RG. These features are listed in Section II-A and further elaborated upon in the reference [64]. This information was utilized for justification performance in a multi-label classification challenge. Although the graders were carefully selected and monitored on their performance on the main labels (RG, NRG), no performance checks were done on the ten additional labels, which may have resulted in less consistent, noisier input data for the training and testing of the models. Future work may be directed at improving the consistency of the additional labels.

This challenge exclusively concentrated on the categorization of CFPs. Examining the impact of incorporating different forms of metadata, which are currently unpublished but accessible in our dataset, on referral performance and justification could be a compelling endeavor. This metadata, while absent for certain photos, contains information such as the camera model, age of the patient, fellow eye, and other anonymous patient identification.

Additionally, we plan to further expand our JustRAIGS dataset by incorporating new, high-quality fundus photographs from diverse screening centers, subject to the appropriate data-use approvals. Moving forward, we aim to include supplemental clinical annotations and metadata where possible, thereby fostering ever more comprehensive and clinically relevant AI solutions in glaucoma screening. In particular, we acknowledge the critical importance of glaucoma stage identification (i.e., distinguishing early, moderate, and advanced disease) as an essential enhancement to the current referability and justification framework. To facilitate such capabilities, future versions of the dataset will emphasize the integration of multimodal clinical data, combining functional assessments such as visual field testing with structural indicators, including OCT-derived metrics. This will enable the development of AI systems not only capable of detecting glaucoma but also of accurately estimating its severity, thereby enhancing clinical applicability.

Future research should prioritize the development of efficient and scalable AI solutions for glaucoma screening. One promising direction involves the use of model compression techniques, such as pruning, quantization, and knowledge distillation, to produce lightweight models that maintain high diagnostic accuracy while enabling real-time inference on portable or low-resource devices. Additionally, adopting federated learning strategies could support training models on distributed datasets across clinical institutions without transferring sensitive patient data. Such approaches promote privacy-preserving, diverse, and clinically robust AI systems that are better suited for deployment in varied healthcare environments.

VII. Conclusion

We showcased the outcomes of community-sourced models that were evaluated using real-world data for the purpose of justifiable referral in glaucoma screening. The top-performing algorithms demonstrated comparable screening capabilities to the meticulously trained and hand-picked human graders, proving their effectiveness in identifying the causes of glaucoma. The top performing algorithms used optic disc localization models as a pre-processing step and vision transformers for the referral task. Additionally, for the justification task, they employed label-

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

smoothing approaches. The dataset we supplied is of an unparalleled scale and reflects real-world screening conditions. The algorithms created with this dataset are expected to contribute significantly to the implementation of reliable AI for justified referral glaucoma screening.

REFERENCES

[1] Y.-C. Tham, X. Li, T. Y. Wong, H. A. Quigley, T. Aung, and C.-Y. Cheng, "Global prevalence of glaucoma and projections of glaucoma burden through 2040: a systematic review and meta-analysis," *Ophthalmology,* vol. 121, no. 11, pp. 2081-2090, 2014.

[2] P. Mokhles, J. S. Schouten, H. J. Beckers, A. Azuara-Blanco, A. Tuulonen, and C. A. Webers, "A systematic review of end-of-life visual impairment in open-angle glaucoma: an epidemiological autopsy," *Journal of glaucoma,* vol. 25, no. 7, pp. 623-628, 2016.

[3] D. Peters, B. Bengtsson, and A. Heijl, "Lifetime risk of blindness in open-angle glaucoma," *American journal of ophthalmology,* vol. 156, no. 4, pp. 724-730, 2013.

[4] R. N. Weinreb, T. Aung, and F. A. Medeiros, "The pathophysiology and treatment of glaucoma: a review," *Jama,* vol. 311, no. 18, pp. 1901-1911, 2014.

[5] Y. Madadi, H. Abu-Serhan, and S. Yousefi, "Domain Adaptation-Based deep learning model for forecasting and diagnosis of glaucoma disease," *Biomedical Signal Processing and Control,* vol. 92, p. 106061, 2024.

[6] Z. Li, Y. He, S. Keel, W. Meng, R. T. Chang, and M. He, "Efficacy of a deep learning system for detecting glaucomatous optic neuropathy based on color fundus photographs," *Ophthalmology,* vol. 125, no. 8, pp. 1199-1206, 2018.

[7] T. W. Rogers *et al.*, "Evaluation of an AI system for the automated detection of glaucoma from stereoscopic optic disc photographs: the European Optic Disc Assessment Study," *Eye,* vol. 33, no. 11, pp. 1791-1797, 2019.

[8] Y. Zhou *et al.*, "A foundation model for generalizable disease detection from retinal images," *Nature,* vol. 622, no. 7981, pp. 156-163, 2023.

[9] A. Thakur, M. Goldbaum, and S. Yousefi, "Predicting glaucoma before onset using deep learning," *Ophthalmology Glaucoma,* vol. 3, no. 4, pp. 262-268, 2020.

[10] Y. Madadi, M. Delsoz, A. S. Khouri, M. Boland, A. Grzybowski, and S. Yousefi, "Applications of artificial intelligence-enabled robots and chatbots in ophthalmology: recent advances and future trends," *Current opinion in ophthalmology,* vol. 35, no. 3, pp. 238-243, 2024.

[11] D. Henson, A. Spencer, R. Harper, and E. Cadman, "Community refinement of glaucoma referrals," *Eye,* vol. 17, no. 1, pp. 21-26, 2003.

[12] Y. J. Song, Y. W. Kim, K. H. Park, Y. K. Kim, H. J. Choi, and J. W. Jeoung, "Comparison of glaucoma patients referred by glaucoma screening versus referral from primary eye clinic," *PLoS One,* vol. 14, no. 1, p. e0210582, 2019.

[13] S. Phene *et al.*, "Deep learning and glaucoma specialists: the relative importance of optic disc features to predict glaucoma referral in fundus photographs," *Ophthalmology,* vol. 126, no. 12, pp. 1627-1639, 2019.

[14] C. De Vente *et al.*, "AIROGS: artificial intelligence for robust glaucoma screening challenge," *IEEE transactions on medical imaging,* 2023.

[15] Z. Zhang *et al.*, "Origa-light: An online retinal fundus image database for glaucoma analysis and research," in *2010 Annual international conference of the IEEE engineering in medicine and biology*, 2010: IEEE, pp. 3065-3068.

[16] F. Fumero, S. Alayón, J. L. Sanchez, J. Sigut, and M. Gonzalez-Hernandez, "RIM-ONE: An open retinal image database for optic nerve evaluation," in *2011 24th international symposium on computer-based medical systems (CBMS)*, 2011: IEEE, pp. 1-6.

[17] J. Odstrcilik *et al.*, "Retinal vessel segmentation by improved matched filtering: evaluation on a new high-resolution fundus image database," *IET Image Processing,* vol. 7, no. 4, pp. 373-383, 2013.

[18] J. Sivaswamy, S. Krishnadas, A. Chakravarty, G. Joshi, and A. S. Tabish, "A comprehensive retinal image dataset for the assessment of glaucoma from the optic nerve head analysis," *JSM Biomedical Imaging Data Papers,* vol. 2, no. 1, p. 1004, 2015.

[19] J. I. Orlando, J. Barbosa Breda, K. Van Keer, M. B. Blaschko, P. J. Blanco, and C. A. Bulant, "Towards a glaucoma risk index based on simulated hemodynamics from fundus images," in *Medical Image Computing and Computer Assisted Intervention–MICCAI 2018: 21st International Conference, Granada, Spain, September 16-20, 2018, Proceedings, Part II 11*, 2018: Springer, pp. 65-73.

[20] J. I. Orlando *et al.*, "Refuge challenge: A unified framework for evaluating automated methods for glaucoma assessment from fundus photographs," *Medical image analysis,* vol. 59, p. 101570, 2020.

[21] H. Fang *et al.*, "Refuge2 challenge: Treasure for multi-domain learning in glaucoma assessment," *arXiv preprint arXiv:2202.08994,* 2022.

[22] J. Wu *et al.*, "Gamma challenge: glaucoma grading from multi-modality images," *Medical Image Analysis,* vol. 90, p. 102938, 2023.

[23] H. G. Lemij, C. de Vente, C. I. Sánchez, and K. A. Vermeer, "Characteristics of a large, labeled data set for the training of artificial intelligence for glaucoma screening with fundus photographs," *Ophthalmology Science,* vol. 3, no. 3, p. 100300, 2023.

[24] N. J. Reus *et al.*, "Clinical assessment of stereoscopic optic disc photographs for glaucoma: the European Optic Disc Assessment Trial," *Ophthalmology,* vol. 117, no. 4, pp. 717-723, 2010.

[25] H. Fu, J. Cheng, Y. Xu, D. W. K. Wong, J. Liu, and X. Cao, "Joint optic disc and cup segmentation based on multi-label deep network and polar transformation," *IEEE transactions on medical imaging,* vol. 37, no. 7, pp. 1597-1605, 2018.

[26] M. A. Kass *et al.*, "The Ocular Hypertension Treatment Study: a randomized trial determines that topical ocular hypotensive medication delays or prevents the onset of primary open-angle glaucoma," *Archives of ophthalmology,* vol. 120, no. 6, pp. 701-713, 2002.

[27] M. O. Gordon and M. A. Kass, *Ocular Hypertension Treatment Study (OHTS): Manual of Procedures*. US Department of Commerce National Technical Information Service, 1995.

[28] M. O. Gordon, M. A. Kass, and O. H. T. S. Group, "The Ocular Hypertension Treatment Study: design and baseline description of the participants," *Archives of Ophthalmology,* vol. 117, no. 5, pp. 573-583, 1999.

[29] E. Kim, S. Lee, G. Kwon, H. Kim, and J. Kim, "Multi-Stage Region-Based Neural Networks for Fine-Grained Glaucoma Screening," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024: IEEE, pp. 1-4.

[30] M. A. G. B. A. Galdran, "Data-Centric Label Smoothing for Explainable Glaucoma Screening from Eye Fundus Images," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 2024, pp. 1-4, doi: 10.1109/ISBI56570.2024.10635220.

[31] D. Presil, "Justraigs Competition Insights: An Ensemble of Methods for Automated Referable Glaucoma Diagnosis," in *2024*

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

IEEE International Symposium on Biomedical Imaging (ISBI), 2024: IEEE, pp. 1-4.

[32] K. Aguilar, V. Cavalcante, C. Carvalho, W. Sabino, and M. Oliveira, "Automatic Glaucoma Classification and Feature Detection for the Justraigs Challenge," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024: IEEE, pp. 1-4.

[33] C. A. a. A. K. K. H. Lin, "Brighteye: Glaucoma Screening with Color Fundus Photographs Based on Vision Transformer," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 2024, pp. 1-4, doi: 10.1109/ISBI56570.2024.10635883.

[34] T. Kubrak, "Automated Detection of Glaucoma and Diagnostic Features for Justraigs Challenge," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024: IEEE, pp. 1-3.

[35] P. Zhang *et al.*, "Detection and Classification of Glaucoma in the Justraigs Challenge: Achievements in Binary and Multilabel Classification," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024: IEEE, pp. 1-4.

[36] Á. Casado-García, J. Heras, M. Ortega, and L. Ramos, "Deep Learning Models for Justified Referral in AI Glaucoma Screening," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024: IEEE, pp. 1-3.

[37] M. Shaurya, A. Varma, and M. Agrawal, "Glaucoma Abnormality Classification Using Yolo and Swin Transformer," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024: IEEE, pp. 1-3.

[38] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "Unet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE transactions on medical imaging,* vol. 39, no. 6, pp. 1856-1867, 2019.

[39] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*, 2019: PMLR, pp. 6105-6114.

[40] Y. Fang *et al.*, "Eva: Exploring the limits of masked visual representation learning at scale," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 19358-19369.

[41] Z. Liu *et al.*, "Swin transformer v2: Scaling up capacity and resolution," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 12009-12019.

[42] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11976-11986.

[43] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2980-2988.

[44] R. Wightman, H. Touvron, and H. Jégou, "Resnet strikes back: An improved training procedure in timm," *arXiv preprint arXiv:2110.00476,* 2021.

[45] S. Woo *et al.*, "Convnext v2: Co-designing and scaling convnets with masked autoencoders," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 16133-16142.

[46] Y. Fang, Q. Sun, X. Wang, T. Huang, X. Wang, and Y. Cao, "Eva-02: A visual representation for neon genesis," *arXiv preprint arXiv:2303.11331,* 2023.

[47] A. M. Reza, "Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement," *Journal of VLSI signal processing systems for signal, image and video technology,* vol. 38, pp. 35-44, 2004.

[48] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556,* 2014.

[49] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929,* 2020.

[50] S. M. Pizer *et al.*, "Adaptive histogram equalization and its variations," *Computer vision, graphics, and image processing,* vol. 39, no. 3, pp. 355-368, 1987.

[51] E. D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, and Q. V. Le, "Autoaugment: Learning augmentation policies from data," *arXiv preprint arXiv:1805.09501,* 2018.

[52] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," *Advances in neural information processing systems,* vol. 34, pp. 12077-12090, 2021.

[53] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770-778.

[54] H. Wang, Z. Li, and W. Zhang, "Get the Best of Both Worlds: Improving Accuracy and Transferability by Grassmann Class Representation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 22478-22487.

[55] Y. Li *et al.*, "Automated Detection of Myopic Maculopathy in MMAC 2023: Achievements in Classification, Segmentation, and Spherical Equivalent Prediction," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2023: Springer, pp. 1-17.

[56] M. E. H. Daho *et al.*, "DISCOVER: 2-d multiview summarization of optical coherence tomography angiography for automatic diabetic retinopathy diagnosis," *Artificial Intelligence in Medicine,* vol. 149, p. 102803, 2024.

[57] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 10012-10022.

[58] R. Wang, T. Lei, R. Cui, B. Zhang, H. Meng, and A. K. Nandi, "Medical image segmentation using deep learning: A survey," *IET image processing,* vol. 16, no. 5, pp. 1243-1267, 2022.

[59] J. Li, J. Chen, Y. Tang, C. Wang, B. A. Landman, and S. K. Zhou, "Transforming medical imaging with Transformers? A comparative review of key properties, current progresses, and future perspectives," *Medical image analysis,* vol. 85, p. 102762, 2023.

[60] E. E. Hwang, D. Chen, Y. Han, L. Jia, and J. Shan, "Multi-Dataset Comparison of Vision Transformers and Convolutional Neural Networks for Detecting Glaucomatous Optic Neuropathy from Fundus Photographs," *Bioengineering,* vol. 10, no. 11, p. 1266, 2023.

[61] Y. Tang *et al.*, "Self-supervised pre-training of swin transformers for 3d medical image analysis," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 20730-20740.

[62] H. Cao *et al.*, "Swin-unet: Unet-like pure transformer for medical image segmentation," in *European conference on computer vision*, 2022: Springer, pp. 205-218.

[63] A. Hatamizadeh *et al.*, "Unetr: Transformers for 3d medical image segmentation," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2022, pp. 574-584.

[64] H. G. Lemij, C. de Vente, C. Sánchez, J. Cuadros, N. Jaccard, and K. Vermeer, "Glaucomatous features in fundus photographs of eyes with 'referable glaucoma'of a large population based labeled data set for training an artificial intelligence (AI) algorithm for

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Transactions on Medical Imaging. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMI.2025.3596874

glaucoma screening," *Investigative Ophthalmology & Visual Science,* vol. 63, no. 7, pp. 2041–A0482-2041–A0482, 2022.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/